ANALYSTLAB AFRICA

Machine Learning Internship Programme

Week 1 — Machine Learning Proposal

1. Problem Framing

Item	Detail
Business Problem	Predict at-risk customers of cancellation to enable proactive retention
Target Variable	Churn (Yes / No)
Problem Type	Binary Classification
Input Variables	19 features: 3 numerical (tenure, MonthlyCharges, TotalCharges) + 16 categorical
Excluded Variable	customerID (unique identifier, no predictive value, risk of data leakage)

2. Preprocessing Strategy

Step	Reason
Removal of customerID	Unique identifier with no predictive signal, risk of overfitting
Correction of TotalCharges type + imputation of 11 missing values	Incorrectly typed column (text); imputation with 0 consistent with tenure = 0
Binary encoding (Partner, Dependents, PhoneService, PaperlessBilling, gender, Churn)	Yes/No variables → 0/1
One-hot encoding (InternetService, Contract, PaymentMethod, MultipleLines, OnlineSecurity,	Categorical variables with 3+ categories, non-ordinal

etc.)	
Scaling (tenure, MonthlyCharges, TotalCharges)	Necessary for scale-sensitive models (logistic regression, KNN, SVM)
Stratified train/test split on Churn	Preserve the 73.5% / 26.5% distribution in both sets
Handling class imbalance	class_weight='balanced' or oversampling (SMOTE) on the training set only

3. Main takeaways from Exploratory Data Analysis (EDA)

The complete exploratory analysis (3 bar charts, 2 histograms, 1 correlation heatmap) is available in the attached Jupyter notebook. Main takeaways:

- "Month-to-month" contracts churn significantly more than 1 or 2-year contracts — commitment duration is a major churn factor.
- Fiber optic customers churn more than DSL or no-internet customers.
- Payment by Electronic check is associated with a higher churn rate than automated payments (bank transfer, credit card).
- The distribution of tenure is bimodal: many very recent customers (0–5 months) and a second group of loyal customers (70+ months).
- tenure is the numerical variable most correlated (negatively) with churn: the longer a customer stays, the less likely they are to leave.

4. Recommended Evaluation Metrics

Since the target is imbalanced (~73.5% No / ~26.5% Yes), accuracy alone would be misleading. Recommended metrics:

- **Recall —** proportion of actual churners correctly identified (priority: do not miss an at-risk customer).
- **Precision —** proportion of "churn" predictions that are actually correct (avoid wasting retention offers).
- **F1-score —** balance between precision and recall.
- **ROC-AUC —** overall ability to separate the two classes, independent of the decision threshold.

5. Recommended Algorithms

Algorithm	Justification
Logistic Regression (baseline)	Simple, fast, interpretable — helps explain churn factors to the client
Random Forest	Captures non-linear interactions, provides feature importance
Gradient Boosting (XGBoost / LightGBM)	Generally the best performing on tabular data of this type
KNN / SVM (optional)	Useful as a point of comparison, less common in production for this use case

6. Recommended Approach

1) Start with a Logistic Regression as an interpretable baseline.

2) Compare with a Random Forest and a Gradient Boosting model.

3) Select the final model based on F1-score and ROC-AUC (not accuracy alone), evaluated on a stratified and unseen test set during training.

4) Document feature importance to present actionable churn factors to the client.